ITACA: principle of operation
Triggered electron and ion readout

Working design note

9 November 2026

Abstract

ITACA measures the electrons and positive ions produced by the same ionising track. Electroluminescence provides the fast energy measurement and electron track. An accepted energy trigger predicts where and when to collect the ion track, positions the movable detector and controls a cathode gate. This note organises the operating sequence, signal formation, readout, signal treatment and reconstruction of both tracks. The new design uses a configurable drift field of 300 V/cm at 15 bar and records the CSA output with a 12-bit ADC every 100 microseconds. Digital reconstruction is a separate stage. The operating choices are design inputs; the ADC, trigger-window prediction and nonlinear inverse described here are the next implementation. Existing productions retain their recorded configurations.

Contents

1 Overall view: one event, two tracks

Consider a neutrinoless double-beta event at (X, Y, Z) and time t_0. Its two electrons deposit a total energy near $Q_{\beta\beta} = 2458$ keV through an extended track. Ionisation produces an electron and a positive ion per primary pair. The relevant positive carrier is the molecular ion Xe_2^+, of charge $+e$. Primary scintillation supplies t_0.

Define $z = 0$ at the ELM entrance and $z = L$ at the cathode drift boundary. Electrons from a deposit at z drift through z; its ions drift through $L - z$. The vertex labels the event, while all its deposits determine the arrival interval. In a uniform drift region,

$$t_e(z) = t_0 + \frac{z}{v_e}, \qquad t_i(z) = t_0 + \frac{L - z}{v_i} + \delta_i. \tag{1}$$

Here δ_i is the calibrated travel time through the focusing structure to the pad. Electron readout timing similarly includes the calibrated delay of EL emission and electronics.

The electron track, or *etrack*, is measured first. The ion track, or *itrack*, arrives seconds later with much less diffusion. This delayed coincidence defines the operating sequence [**?**]:

1. Record primary scintillation and read the EL signal. Sum its calibrated response to estimate event energy E_{fast}.

2. Accept the energy trigger if

$$|E_{\text{fast}} - Q_{\beta\beta}| \leq \text{FWHM}_E(Q_{\beta\beta}). \tag{2}$$

   The ROI is ± 1 FWHM, with full width 2 FWHM. Its width is expressed in energy units and calibrated for the fast energy estimator.

3. For accepted events, the fast-trigger algorithm (FTA) estimates the etrack barycentre, longitudinal and transverse extent and arrival-time distribution. Together with t_0, these predict the ion footprint and arrival window. Under the MARS design, the movable ion detector is positioned to contain that footprint, including its uncertainty.

4. Acquire the CSA baseline, open the cathode gate shortly before the predicted first ions, collect the track, close the gate after its predicted last ions and finish a short post-arrival recording. Rejected candidates leave the ion gate closed.

The EL measurement supplies the primary energy estimate; the ion branch is optimised for topology. Both branches yield reconstructed charge and position for consistency checks and combined track analysis. The manuscript also describes a barrel fibre detector for scintillation timing and calorimetry. The selected fast-energy channel, whether that detector or the calibrated ELM sum, and its FWHM belong to the trigger configuration.

1.1 Operating defaults

Quantity	New operating proposal
Chamber length; pressure; gas	1.5 m; 15 bar; Xe–15% He
Drift field E_d	300 V/cm
Reduced drift field E_d/P	20 V/cm/bar, derived
Electron drift velocity v_e	Approximately 0.6 mm/μs; graph estimate
Maximum electron drift time	Approximately 2.5 ms
Ion velocity model	$v_i = \mu_i E_d$, calibrated mobility
Ion pitch	4 mm initially; configurable
EL pitch; hole diameter; depth	4 mm; 2.5 mm; 5 mm
EL reduced field	2 kV/cm/bar initially
EL time buckets	250 ns
Ion ADC	100 μs; 12 bits; provisional 0–1.8 V
Digital voltage averaging	5 ms initially
Stored ADC samples	Compressed HDF5, `UInt16`
Energy ROI	$Q_{\beta\beta} \pm 1$ FWHM
Gate opening advance	1 ms before the predicted early boundary

Table 1: User design choices agreed on 9 November 2026. Cards select geometry, transport, acquisition and reconstruction independently.

Drift field and pressure are independent card inputs; their ratio is derived. An explicitly supplied ratio must agree with them. The EL reduced field is a different parameter applying inside the holes.

At 300 V/cm and 15 bar, $E_d/P = 20$ V/cm/bar. Figure 8 of Ref. [**?**] gives approximately $v_e = 0.6$ mm/μs for Xe–15% He. This is a graphical estimate transferred by reduced-field scaling from measurements at 1–9 bar, rather than a direct 15-bar calibration. Therefore $L/v_e \simeq 2.5$ ms is the illustrative maximum drift time. EL formation and electronics add their response duration. A refined transport calculation or measurement will supply the gas-, field- and temperature-dependent card value.

The previous ion design used 0.1 m/s at 200 V/cm. Keeping that mobility gives $\mu_i = 5 \times 10^{-6}$ m^2/(V s) and, illustratively, $v_i = 0.15$ m/s at 300 V/cm. Maximum ion drift then takes 10 s; a 10–100 mm longitudinal extent gives approximately 67–667 ms of arrivals. This assumes field-independent mobility at fixed gas and pressure and requires calibration. One second is an event-duration scale, not a fixed acquisition length.

A reconstructed time bucket T_i corresponds to $v_i T_i$ along z. At this illustrative velocity, 40 ms represents 6 mm and 4 mm corresponds to 26.7 ms. ADC sampling, reconstructed bucket width

and transverse pitch are separate choices. Likewise 250 ns corresponds to approximately 0.15 mm for the new electron velocity. Earlier productions retain their original velocity and distance-to-time conversions.

1.2 Predicting when ion acquisition ends

For inferred undiffused longitudinal extent $[z_{\min}, z_{\max}]$, the nominal collection boundaries are

$$t_{is} = t_0 + \frac{L - z_{\max}}{v_i} + \delta_i, \qquad t_{ie} = t_0 + \frac{L - z_{\min}}{v_i} + \delta_i. \tag{3}$$

The earliest electron deposits correspond to the latest ions. The FTA derives these boundaries from EL measurements and calibration. Simulation truth evaluates their accuracy separately.

Diffusion is statistical. For independent, zero-mean displacements, a useful variance relation is

$$s_e^2 = s_0^2 + \langle \sigma_{e,L}^2(z) \rangle + s_{\text{readout}}^2, \qquad s_i^2 = s_0^2 + \langle \sigma_{i,L}^2(L - z) \rangle. \tag{4}$$

All terms are expressed in spatial units and consistently charge-weighted, correcting attachment and channel response. These relations guide a fit to the longitudinal profile; its endpoints have uncertainties. EL emission, sensor response and sampling enter s_{readout}. The predicted interval is defined by a specified retained-charge fraction and timing uncertainty.

The gate opens at $t_{is} - m_{\text{early}}$ and closes at $t_{ie} + m_{\text{late}}$ after converting pad-arrival predictions to gate-crossing times. The early margin includes the agreed 1 ms advance and uncertainty allowance; the end margin covers the late diffusion tail and timing error. ADC recording starts before opening and ends after the last admitted ions have crossed the collection structure and the digital filter has finished. The short post-arrival recording measures the final CSA state, which still contains accumulated charge.

A gap or weak section of the itrack does not itself end acquisition. Validation measures excluded charge and endpoint errors as margins are varied. The three-second post-event recording used in earlier CSA studies remains useful for selected recovery tests. Ordinary acquisitions use the event-specific window and short filter-completion interval.

2 EL signal formation and readout

Primary-pair sampling is shared between arms. For $\overline{N} = E/W$, the Fano prescription is $\mathrm{Var}(N) = F\overline{N}$. The same realised count feeds ions and electrons. Integer sampling at small means preserves the mean and respects the variance allowed for integer counts. Electrons drift, diffuse and survive attachment with probability $\exp[-t_e/\tau_e]$. The electron lifetime is configurable, initially 15 ms. Ions receive no electron-attachment loss.

The ELM focuses electrons into holes. The current optical model assumes unit electron collection for EL reduced fields at least 2 kV/cm/bar. Mean yield per entering electron is parameterised as

$$Y_g = a_g P d[(E_{\text{EL}}/P) - b_g], \tag{5}$$

with d in cm, P in bar and reduced field in kV/cm/bar. Gas-dependent coefficients and their provenance are documented in `../../docs/elm.tex` and selected by the detector card. At fixed entering count N_e,

$$\langle N_{\text{EL}} \rangle = N_e Y_g, \qquad \mathrm{Var}(N_{\text{EL}}) = N_e J Y_g. \tag{6}$$

The EL variance prescription follows Oliveira et al. [**?**]. Values $J = 0.030, 0.015, 0.025$ at reduced fields $2, 2.5, 3$ are approximate user readings of their figure, adopted for both gases.

For uniform axial isotropic emission in an absorbing cylindrical hole of radius r and depth d, the direct exit acceptance is

$$A_{\mathrm{exit}} = \frac{1}{2}\left[1 - \frac{\sqrt{d^2 + r^2} - r}{d}\right]. \tag{7}$$

This is the average forward solid angle of the exit disk. Photons hitting the walls are absorbed. TPB at the guide entrance converts VUV to blue with the adopted probability 0.65; isotropic emission sends half into the guide. In the meridional capture approximation, $f_{\mathrm{trap}|+} = 1 - n_x/n_g$ for guide and surrounding gas indices n_g and n_x. Propagation, electrode transmission, coupling and coverage multiply these factors. The guides have a 0.5 mm radial xenon gap and are coupled to the SiPM with Norland NOA 87. Packaged PDE includes the sensor window. Optical boundary conditions and derivations are in the ELM note.

The detector stage saves blue photons reaching each SiPM per time bucket. Sensor response applies PDE, microcell recovery and dark activity; electronics then adds its transfer, noise and finite range. The present EL readout model represents charge per bucket in PE equivalents, with a provisional current limit. An EL ADC model requires its own device specification. Appendix **??** describes sensor and readout details.

3 Ion collection and CSA–ADC readout

An accepted trigger positions MARS and controls an electrostatic gate ahead of the ion focusing grid (IFG). As in the manuscript, pads are near ground, the IFG face nearer them is at 3 kV and the TPC-facing surface at 8 kV. A gate at 7.8–7.9 kV establishes a barrier to positive ions approaching the 8 kV face. Raising it to approximately 8 kV opens the selected window. Transmission and transfer delay depend on the electrode geometry. Figure **??** shows the collection structure.

The closed gate prevents rejected charge from reaching the CSA. Interception and disposal of blocked ions belong to the gate transport design, so stored ions are not subsequently counted as a new accepted event. Fixed-potential electrodes shield the pads from switching; residual aperture coupling can be bounded by the induced pad charge per unit gate-voltage change. Its magnitude requires that geometry calculation.

For a 10 mm IFG and 5 mm pad gap, the manuscript's 8/3/0 kV potentials give nominal fields of 5 and 6 kV/cm respectively. The ion collection geometry and potentials are retained from the manuscript. At 300 V/cm over 150 cm, the drift voltage difference is 45 kV and the anode-side drift boundary is 53 kV. An extra transfer gap adds its specified voltage. At 2 kV/cm/bar, 15 bar and 5 mm EL depth, the EL voltage difference is 15 kV. The guide grading length is derived from the resulting exit potential and selected grading field.

Positive ions induce pad current during their final approach. The weighting potential ϕ_w is defined with the selected pad at unit potential and all other conductors at zero. The biased transport field is a different problem. For a positive ion, in the adopted input-current sign convention,

$$i(t) = e\frac{\mathrm{d}\phi_w(\mathbf{r}(t))}{\mathrm{d}t}, \qquad \int i\,\mathrm{d}t = e[\phi_w(\mathbf{r}_f) - \phi_w(\mathbf{r}_i)]. \tag{8}$$

Finite-path charge is preserved and the current ends at collection. Appendix **??** gives the benchmark pad model and circuit schematics.

The CSA integrates input current and continuously discharges through its feedback. Its output, including analogue noise, is digitised every $\Delta t_s = 100$ μs. Initially the ADC has 12 bits and a 0–1.8 V range; 14 bits and 10 μs provide comparison settings. Define $\Delta V = (V_{\max} - V_{\min})/2^B$. An explicit ideal code convention is

$$c_k = \mathrm{clip}\left(\left\lfloor \frac{V_k - V_{\min}}{\Delta V} \right\rfloor, 0, 2^B - 1\right). \tag{9}$$

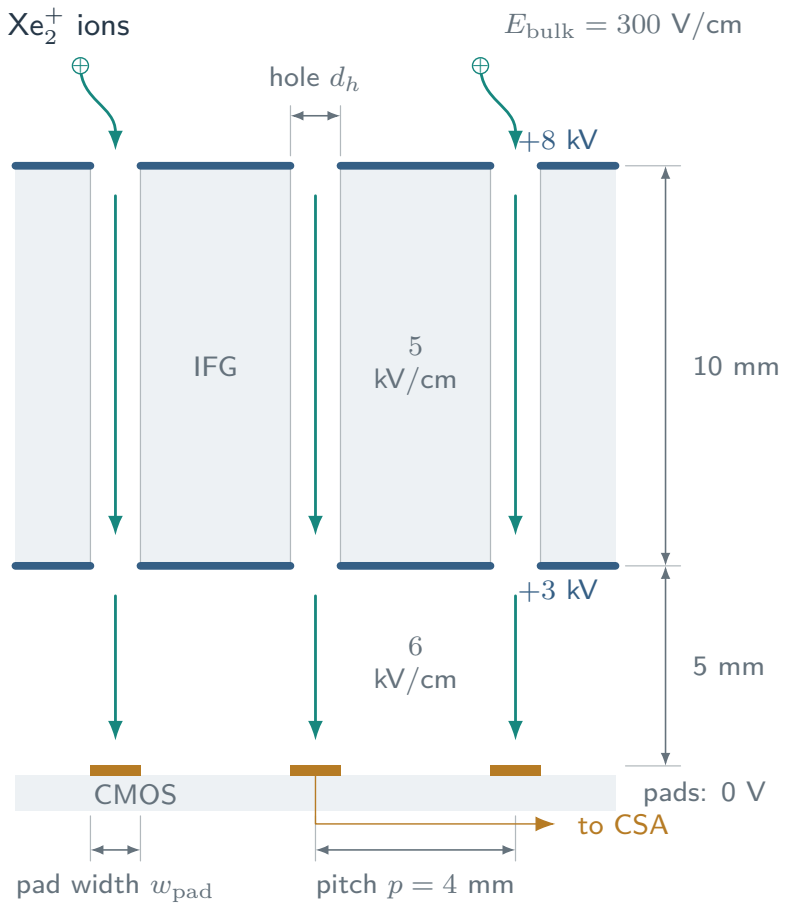


Figure 1: Ion collection geometry, adapted from the manuscript. The 10 mm IFG and 5 mm pad gap are drawn in their 2:1 vertical proportion; transverse features are schematic. Potentials remain 8/3/0 kV. The gate lies upstream of the drawn IFG.

Interior codes can be associated with bin-centre voltages. Simulation records out-of-range truth separately; the reconstruction interprets endpoint codes as potentially saturated observations. Transition and endpoint tests specify this convention.

The device stage ends at codes, timing and calibration metadata. Arrays use `UInt16` and compressed HDF5. ADC noise is specified separately from CSA noise. Analogue input bandwidth and the treatment of noise above Nyquist are explicit: spectral power is either suppressed by the analogue transfer or contributes through aliasing. Digital filtering follows digitisation.

4 Signal treatment

4.1 EL channel calibration and SiPM correction

Subtract the measured pedestal and expected dark contribution, calibrate single-avalanche charge and account for channel timing and response. The sensor correction maps measured charge to primary photodetections using its calibrated recovery model and preceding measured history. PDE and optical calibration relate that estimate to ionisation charge. Attachment is corrected using measured drift time and lifetime; residual fluctuations remain.

Sensor compression is invertible where the calibrated response retains sensitivity. Electronics clipping provides a bound rather than the original charge. Reconstruction records this distinction. The fast energy estimator uses corrections compatible with trigger latency; offline processing may refine them. Each estimator has its own calibrated resolution.

4.2 Ion baseline, filtering and linear reconstruction

Pre-arrival ADC samples estimate the starting CSA state and its uncertainty. An initial 5 ms digital average combines 50 samples. Filter timing, boundary treatment and gain belong to the calibration. Filtering can work directly in ADC units, with gain expressed in counts per ion.

For an ideal linear CSA, let $u = V - V_0$ be displacement from its resting voltage:

$$C_f \dot{u} + \frac{u}{R_f} = -i(t), \qquad \tau_f = R_f C_f. \tag{10}$$

The input charge in a reconstructed bucket is

$$Q_n = -C_f \left[u(t_n) - u(t_{n-1}) + \frac{1}{\tau_f} \int_{t_{n-1}}^{t_n} u(t)\,\mathrm{d}t \right]. \tag{11}$$

Measured gain and bandwidth replace the ideal response. Induction and filtering distribute a deposit between neighbouring buckets; a calibrated response matrix removes that mixing. Noise covariance passes through the same operation. Signed reconstructed charge is retained.

4.3 Nonlinear ion reconstruction

The existing nonlinear forward model follows $x = V_0 - V$ with

$$\dot{x} = g(x)\frac{i(t)}{e} - r(x), \tag{12}$$

where g is voltage gain per elementary charge and r the recovery rate. SPICE calibration uses disjoint fitting and validation histories. This model predicts voltage; the nonlinear inverse is a separate operation.

The proposed inverse estimates bucket charges and initial state by fitting the calibrated response to measured ADC history, including noise and quantisation. The current prescription within each charge bucket is explicit. Linear and nonlinear estimators receive the same codes. MC truth is reserved for validation. Any automatic model choice depends on measured response and the calibration domain. Bias, uncertainty, saturation and fit quality are reported separately.

5 Reconstruction of energy and position

5.1 Electron track

Channel coordinates provide transverse positions; corrected electron timing gives $z = v_e(t_e - t_0)$. The measured distribution has resolution set by diffusion, hole pitch and signal response. For corrected photodetection estimates $\widehat{n}_{cb}$, a schematic energy estimator is

$$\widehat{E}_e = K_E \sum_{c,b} \frac{\widehat{n}_{cb}}{\epsilon_{cb}} \exp(t_{e,b}/\tau_e), \tag{13}$$

where ϵ_{cb} is the relative channel/spatial response and K_E the absolute energy calibration. Corrections already included in $\widehat{n}$ are applied once. Acceptance and saturation information accompany the sum.

5.2 Ion track and combined observables

Pad coordinates and reconstructed collection times give $z_i = L - v_i(t_i - t_0 - \delta_i)$. The measured MARS position transforms local pad coordinates into the ELM's global transverse frame. The ion charge sum gives $\widehat{E}_i = K_i \sum_{p,n} \widehat{N}_{p,n}$ with its propagated covariance. The EL branch remains the primary calorimeter.

Each reconstructed track provides a barycentre, longitudinal/transverse extent, connected components, backbone and endpoint charge. Moments use consistently corrected weights. Pure MC, diffused and discretised reconstructions remain distinct validation levels. Comparing the etrack and itrack in a shared frame tests matching and improves topological information. Selections and positive-pulse displays are specified separately from signed charge sums used for energy.

6 Configuration, storage and validation

Detector cards specify gas, fields, geometry, lifetime, transport and optics. Electronics cards specify CSA or SiPM response and readout. Acquisition cards specify ROI, gate times, channel region and margins. Reconstruction cards specify filters, estimators and output buckets. Outputs carry units, source identities, seeds and calibration hashes.

The saved stages are ion counts per pad/bucket and photons per SiPM/bucket; device response; and reconstructed charge and position. The new ion device output is ADC codes. Acquisition selects channels from the EL-predicted detector region or an explicit hardware channel set. Truth-selected pixels from previous simulations supply a storage benchmark, not a hardware selection algorithm.

At 100 μs, each 40 ms interval has 400 samples and occupies 800 bytes per channel before compression. For N_p channels recorded for T seconds, payload is approximately $2N_pT/\Delta t_s$ bytes. The short post-arrival interval reduces storage without waiting for physical feedback recovery. Pilot measurements establish compression and window sizes before production.

Checks cover charge conservation, ADC transitions and range, reproducibility, noise spectrum/covariance, temporal convergence and independent SPICE waveforms. Reconstruction tests compare linear and nonlinear charge recovery versus intensity, history, timing and saturation.

Trigger tests compare EL-predicted windows with actual ion arrivals and measure excluded charge. A successive-event test carries the remaining CSA state between accepted events: gating blocks rejected charge but does not reset admitted charge.

Transport and collection live in `src/event_analysis`; electronics in `src/ion_signal`. Existing implementations are documented in `csa_design.tex`, `../../docs/ion_csa_nonlinear_model.html`, and `../../docs/sipm_statistical_implementation.md`. Julia calculates and saves numerical results to HDF5; Python generates report figures and tables. This note defines the new operating model before implementation.

A CSA geometry, circuit and response details

Figure **??** shows the equivalent pixel circuit and ADC readout. Figure **??** gives the transistor schematic supplied by Rogelio Palomo. The calibrated 180 nm circuit uses a 1.8 V supply,

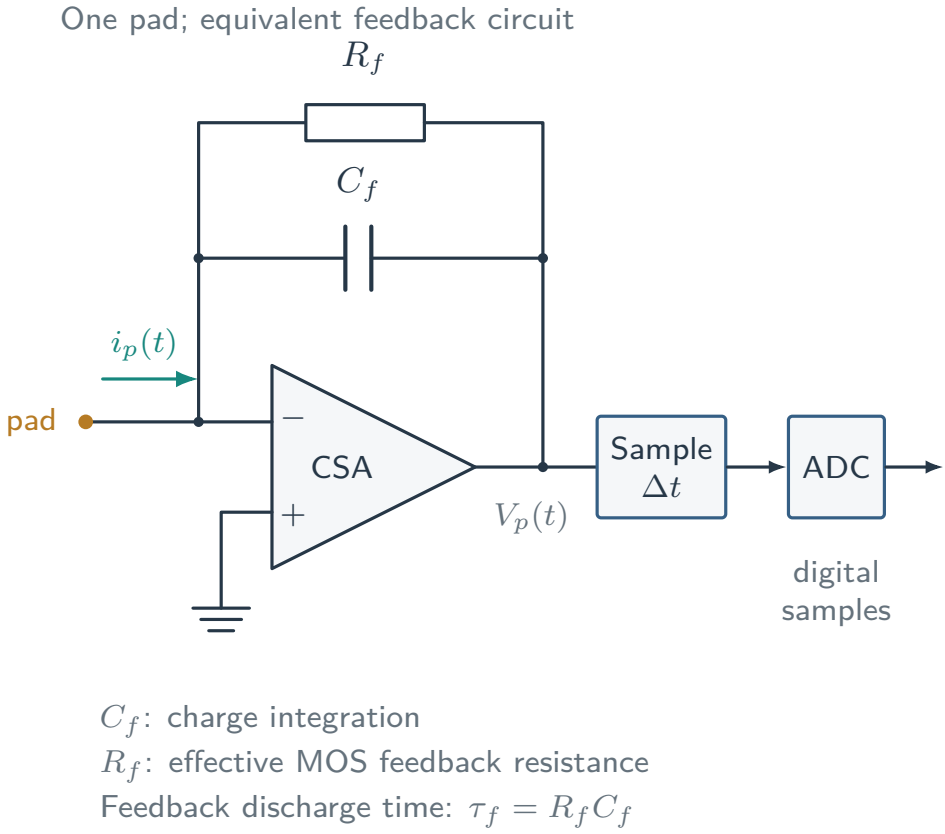


Figure 2: Equivalent CSA circuit and digitised readout, adapted from the manuscript. The effective resistance represents the MOS feedback discharge.

$C_f = 3$ fF and external input capacitance 1.25 pF. Its resting output is approximately 0.886 V and small-signal decay time approximately 19.5 s. Dimensions and biases are in `csa_design.tex` and its generated tables. A circuit change requires revalidation.

For the square-pad benchmark of half-width a in a grounded plane,

$$\phi_w(z) = \frac{2}{\pi}\tan^{-1}\frac{a^2}{z\sqrt{z^2+2a^2}}, \qquad W(z) = -\phi_w'(z) = \frac{4a^2}{\pi(z^2+a^2)\sqrt{z^2+2a^2}}. \tag{14}$$

An ion approaching at speed v, collected at $t = 0$, gives $i_1(t) = evW(-vt)$ over $-g/v \le t < 0$. Its integral is $e[1-\phi_w(g)]$, following the weighting-field prescription [**?**]. The manuscript contains the

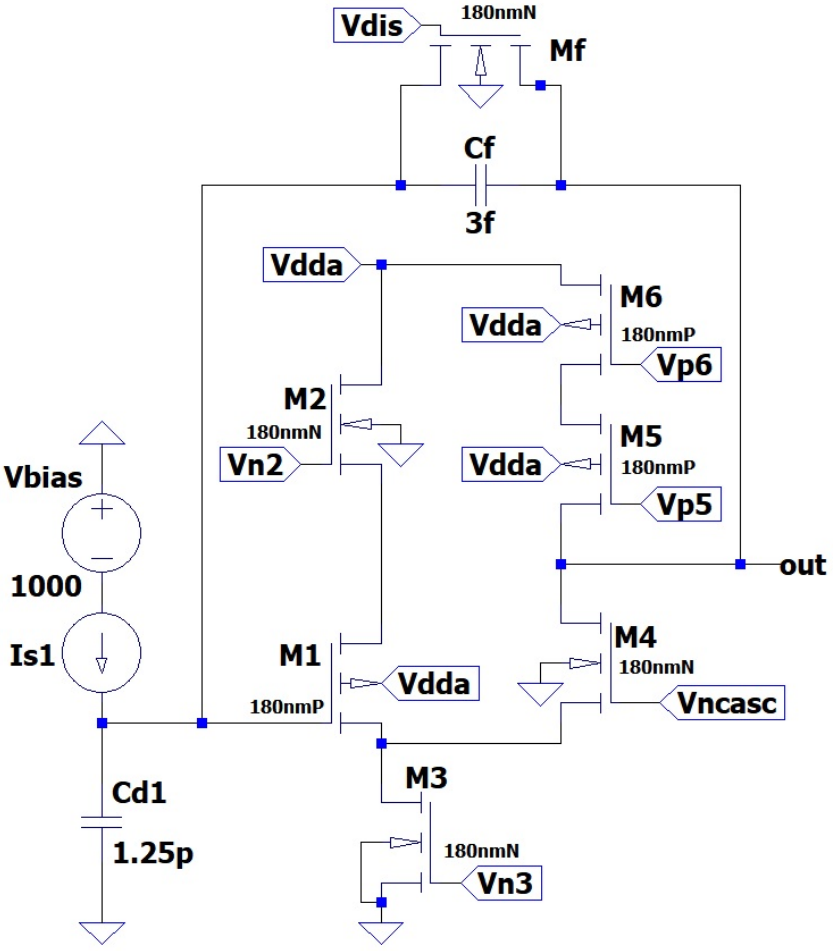


Figure 3: CSA transistor schematic reused from the ITACA manuscript and CSA note. The pad drives the input transistor; C_f integrates charge and the feedback transistor supplies the discharge path.

square-pad derivation. The actual perforated electrodes and dielectrics define the eventual detector weighting solution.

For one-sided output-noise spectrum $S_V(f)$ and linear measurement filter $H(f)$, variance is $\int S_V(f)|H(f)|^2 \mathrm{d}f$. Earlier calculations combine averaging and frame subtraction before sampling reconstructed noise. The new ADC model samples voltage noise before quantisation, then applies digital reconstruction. Compression can retain information; actual saturation loses it. The inverse must distinguish these behaviours.

B SiPM microcells and readout details

The current optical design illuminates a centred 2.5 mm disk on a 3×3 mm^2 Hamamatsu S14160-3015PS with 15 μm cells. Its card uses 39,984 total and 21,808 illuminated cells, gain 3.6×10^5 and PDE 0.32 as a provisional blue-spectrum average [**?**]. The 12.5 ns exponential recovery constant is provisional, with 25 ns as a sensitivity case.

Photodetections are binomial samples of incident photons with probability PDE. A cell firing Δt after its preceding avalanche contributes

$$q = q_0[1 - \exp(-\Delta t/\tau_s)], \qquad q_0 = eG. \tag{15}$$

Its state persists between buckets. Dark triggers also change recovery; the current card takes a 700 kHz total dark rate at its reference operating point. Crosstalk, afterpulsing and gain spread are disabled in that baseline.

The statistical model carries the sum of recovery fractions and their squares. Sparse buckets use individual trigger times; denser ones use substeps and finite-cell occupancy moments. Illuminated and dark-only cell populations retain their identities. The implementation and reference tests are described in `../../docs/sipm_statistical_implementation.md`.

Under steady uniform illumination of M participating cells, expected trigger count n over duration T gives mean PE-equivalent charge

$$m = \frac{n}{1 + kn}, \qquad k = \frac{\tau_s}{MT}, \qquad n = \frac{m}{1 - km}. \tag{16}$$

This AXEL-related mean correction [**?**] is a limiting case and calibration check. Transient correction uses preceding measured history. The inverse becomes poorly conditioned near the asymptote.

The provisional readout converts charge to bucket-average current $I = Q/T$, adds 0.5 PE-equivalent RMS independent electronics noise per 250 ns bucket and limits output at 1 mA. These are card-selected assumptions. A change of bucket duration requires consistent electronics noise and transfer settings. Dark fluctuations are separate from electronics noise. Sensor charge saved before electronics permits that stage to be regenerated independently.